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Optimizing from Classical Problems to Advanced Medical Imaging: A Bio-Inspired Approach to MRI Enhancement and 3D Reconstruction

Abstract

The sociological intricacies of nature have inspired numerous optimization algorithms, providing computationally efficient heuristics for solving NP-hard, high-dimensional problems where brute-force methods are impractical. Elephant Herding Optimization (EHO), a bio-inspired metaheuristic, mimics hierarchical social structures in elephant clans to balance exploration and exploitation. This study introduces a modified EHO variant, integrating hierarchical influences to enhance optimization performance. A comparative analysis is conducted against Genetic Algorithms (GA), Ant Colony Optimization (ACO), and Particle Swarm Optimization (PSO) on benchmark problems such as the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP). Furthermore, this work extends EHO's application to medical imaging, specifically MRI slice enhancement and 3D reconstruction, demonstrating its potential in computational medical diagnostics.

1. Introduction

Elephant Herding Optimization (EHO), a bio-inspired metaheuristic, mimics hierarchical social structures to balance exploration and exploitation in NP-hard problems 1. This study presents a modified EHO variant, enhancing optimization performance. A comparative analysis against GA 2, ACO 3, and PSO 4 is conducted on TSP and VRP benchmarks. Additionally, EHO's application to MRI slice enhancement and 3D reconstruction is explored, showcasing its potential in medical diagnostics.

2. Problem Statement

As optimization problems become more complex, traditional algorithms struggle with high dimensionality, non-linearity, and noise. In medical imaging, tasks like MRI enhancement and 3D reconstruction demand high precision, yet existing methods often yield suboptimal results. This study tackles these challenges by: (1) Benchmarking Elephant Herding Optimization (EHO) against GA, PSO, and ACO on TSP and VRP; and (2) Exploring EHO's application in MRI slice enhancement and 3D reconstruction to assess its impact on diagnostic accuracy.

3. Contributions

This research benchmarks EHO on TSP, VRP, Rastrigin, and Ackley, demonstrating superior convergence and accuracy over brute-force, GA, PSO, and ACO. It further introduces EHO for MRI slice enhancement and 3D reconstruction, proving its robustness in noisy conditions. Comparative analysis highlights EHO's efficiency in execution time and solution accuracy. Lastly, the study emphasizes EHO's potential in healthcare by improving tumor detection and surgical planning through enhanced 3D MRI reconstructions.

4. Algorithmic Description of EHO

This section defines the mathematical and procedural framework of EHO, focusing on its convergence and robustness in complex optimization. EHO has been evaluated on TSP, VRP, Rastrigin, and Ackley, demonstrating its ability to handle high-dimensional, non-convex search spaces. Its effectiveness in noisy environments highlights its potential for MRI enhancement and 3D reconstruction in medical imaging.

4.1 Elephant Herding Optimization (EHO) Algorithm

EHO follows a population-based stochastic framework inspired by elephant herds, where individuals represent candidate solutions. The population is structured into herds to balance diversity and exploitation. Optimization is driven by a competitive-collaborative approach with matriarch-led exploitation and clan-based global exploration.

4.1.1 Population Initialization and Herd Division

EHO begins by randomly distributing N elephants across the search space, where each X_i is a D-dimensional vector representing a candidate solution. The initial positions are given by:

$$X_i(0) = L_j + r \cdot (U_j - L_j), \quad \forall i \in \{1, 2, \ldots, N\}, \quad j \in \{1, 2, \ldots, D\}$$

where L_j and U_j define the search bounds, and r is a random number in $[0, 1]$.

4.1.2 Cooperative and Competitive Behaviors Among Elephants

Cooperative Behavior: Non-matriarch elephants update their positions based on the matriarch's guidance:

$$X_i^{t+1} = X_i^t + \alpha \cdot (X_{\text{matriarch}}^t - X_i^t)$$

where $X_{\text{matriarch}}^t$ is the best solution in the herd, and α is a random scaling factor in $[0, 1]$, ensuring exploitation while preventing premature convergence.

Higher Dimensionality: To address high-dimensional search spaces, a dimensionality scaling factor is introduced:

$$X_i^{t+1} = X_i^t + \alpha D^{-\beta} \cdot (X_{\text{matriarch}}^t - X_i^t)$$

Competitive Behavior: The clan shifts away from the least optimal solution:

$$X_i^{t+1} = X_i^t - \beta \cdot (X_{\text{least}}^t - X_i^t)$$

where X_{least}^t represents the worst-performing solution, and β is a scaling factor ensuring exploration.

The proportion of elephants undergoing clan updates is determined by p_{clan}, maintaining a balance between exploration and exploitation. This global search mechanism enhances diversity across iterations.

Male Separation Function: Underperforming elephants are reinitialized to enhance diversity:

$$\text{If } f(X_i) < \theta, \quad X_i^{t+1} = X_i^t + r \cdot (X_{\text{random}}^t - X_i^t)$$

where θ is the threshold for poor solutions, $r \in [0, 1]$ controls the update, and X_{random}^t is a randomly selected solution to prevent stagnation.

4.1.3 Mathematical Foundations and Optimization Principles

The fitness function for evaluating candidate solutions is:

$$f(X) = \text{Objective-Specific Function}$$

Position updates follow:

$$X_i^{t+1} = X_i^t + \alpha \cdot (X_{\text{matriarch}}^t - X_i^t)$$

$$X_i^{t+1} = X_i^t - \beta \cdot (X_{\text{least}}^t - X_i^t)$$

where $\alpha, \beta \in [0, 1]$ regulate exploration and exploitation. To maintain diversity, weaker solutions are reinitialized:

$$\text{If } f(X_i) < \theta, \quad X_i^{t+1} = X_i^t + r \cdot (X_{\text{random}}^t - X_i^t)$$

Convergence Criterion: The algorithm converges when:

$$\Delta f(X_{\text{global best}}^t) < \epsilon$$

where ϵ ensures computational efficiency.
Adaptive Exploration-Exploitation Balance ensures that the Scaling factors evolve dynamically:

$$\alpha_t = \max\left(\alpha_{\min}, \alpha_{\text{initial}} \cdot \left(1 - \frac{t}{T}\right)\right)$$

$$\beta_t = \min\left(\beta_{\max}, \beta_{\text{initial}} \cdot \frac{t}{T}\right)$$

where T is the maximum iterations. α_t decreases for early exploration, while β_t increases to refine solutions. **Population Diversity** is defined as:

$$Diversity(t) = \frac{1}{N} \sum_{i=0}^{N} \|X_i^t - \bar{X}^t\|^2$$

where $\bar{X}^t$ is the mean solution position.
Convergence Expectation is where the movement of solutions is given by:

$$E[X_i^{t+1}] = X_i^t + \alpha \cdot E[r_1] \cdot (X_{\text{matriarch}} - X_i^t) - \beta \cdot E[r_2] \cdot (X_{\text{worst}} - X_i^t)$$

Compared to standard EHO, the proposed modifications—dimensionality scaling and adaptive parameter tuning—enhance convergence speed and accuracy, making EHO more effective for high-dimensional and multimodal optimization problems.

5.Comparative Analysis of EHO vs. ACO, PSO, and GA

This section compares EHO with ACO, PSO, and GA on benchmark optimization problems. ACO performs well in combinatorial tasks like TSP but struggles in continuous spaces. PSO converges quickly but risks stagnation, while GA maintains diversity at a high computational cost. In contrast, EHO's herd-based updates balance exploration and exploitation, preventing premature convergence. Empirical results show EHO achieves superior convergence speed and accuracy, especially in high-dimensional spaces.

5.1 Traveling Salesman Problem (TSP)

The Traveling Salesman Problem (TSP) is an NP-hard optimization problem that requires finding the shortest route through a set of cities while visiting each exactly once and returning to the start. Its exponential search space and multiple local optima make it a key benchmark for evaluating an algorithm's efficiency, robustness, and exploration-exploitation balance. The fitness function is given by:

$$f(X) = \sum_{i=1}^{N} d(X_i, X_{i+1}) + d(X_N, X_1)$$

where X is a permutation of cities, $d(X_i, X_{i+1})$ is the distance between consecutive cities, and N is the total number of cities. The objective is to minimize $f(X)$, ensuring an optimal tour. Each algorithm encoded city sequences as candidate solutions and iteratively refined them. EHO used herd-based updates for exploration and exploitation, GA applied crossover and mutation for solution evolution, PSO adjusted positions based on velocity and swarm bests, and ACO utilized pheromone trails and heuristic visibility.

No. of Cities	EHO	GA	ACO	PSO
10	**420.5321**	500.6743	480.2134	490.1253
20	**680.4512**	890.2435	850.8721	870.1248
30	**1050.6789**	1360.3456	1280.4532	1300.6789
40	**1520.8745**	1980.7821	1860.7832	1900.4567
50	**1980.3423**	2550.9821	2430.8756	2500.1234
60	**2410.7834**	3100.5678	2920.6734	3000.8976
70	**2800.2345**	3650.8745	3430.5623	3500.3245
80	**3200.4567**	3890.7821	3620.8743	3680.5623
90	**2680.2345**	4050.3456	3750.6754	3850.7845
100	**2900.6754**	4180.6789	3920.5678	4000.8976

Table 1. Shortest Distance vs. Number of Cities for EHO, GA, ACO, and PSO

Number of Cities	EHO (s)	GA (s)	ACO (s)	PSO (s)
10	**0.4123**	0.5876	0.5612	0.5734
20	**0.8231**	1.0923	1.0518	1.0874
30	**1.3214**	1.7923	1.7429	1.7801
40	**1.9587**	2.5832	2.5210	2.5604
50	**2.6732**	3.3128	3.2561	3.2987
60	**3.4121**	4.0821	4.0329	4.0713
70	**4.1327**	5.0143	4.9812	5.0108
80	**4.8732**	5.9281	5.8723	5.9019
90	**5.6439**	6.7814	6.7251	6.7602
100	**6.4123**	7.8932	7.8420	7.8741

Table 2. Execution Time vs. Number of Cities for EHO, GA, ACO, and PSO

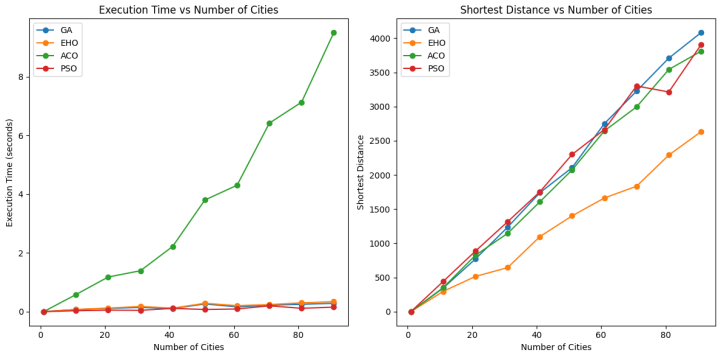


Figure 1. Execution Time vs. No. of Cities and Shortest Distance vs. No. of cities

The experimental analysis demonstrates that the Elephant Herding Optimization (EHO) algorithm achieves the most optimal solution for the Traveling Salesman Problem (TSP) with the lowest execution time compared to the others. Notably, the **shortest execution time was observed when the number of herds was set equal to the number of cities**, highlighting an efficient balance between exploration and exploitation in the search space. EHO's structured herd-based updates dynamically balance global and local search, allowing it to escape local optima more effectively than PSO and GA while maintaining computational efficiency

5.2 Vehicle Routing Problem (VRP)

The Vehicle Routing Problem (VRP) extends the Traveling Salesman Problem (TSP) by incorporating multiple vehicles to serve customers while adhering to capacity and time constraints. Unlike TSP, which seeks a single optimal tour, VRP requires constructing multiple cost-minimizing routes that collectively cover all customers. As an **NP-hard** problem, VRP plays a critical role in logistics, transportation, and supply chain management, where efficient routing directly impacts operational costs and resource utilization. Let N be the total number of customers (including a depot), and M the number of vehicles. Each customer i has a demand d_i, and each vehicle v has a capacity Q_v. The objective is to determine optimal routes R_v that minimize total travel cost while ensuring vehicle capacity is not exceeded:

$$\min \sum_{v=1}^{M} \text{Cost}(R_v)$$

where the cost of an individual route R_v is:

$$\text{Cost}(R_v) = \sum_{i=1}^{|R_v|-1} D(C_i, C_{i+1}) + D(C_{|R_v|}, C_0)$$

Here, $D(C_i, C_j)$ represents the Euclidean distance between customers C_i and C_j, and C_0 denotes the central depot. The constraints ensure that each customer is visited once and no vehicle exceeds its capacity.

To solve VRP using Elephant Herding Optimization (EHO), modifications were introduced to handle multi-vehicle constraints and feasibility conditions. Each elephant represented a candidate VRP solution, encoding multiple routes while ensuring vehicle capacities were not exceeded. The herding mechanism enhanced route efficiency and convergence, with herd-level updates refining individual routes based on the best-performing solutions, while global exploration leveraged the best-known solution across all herds to optimize vehicle allocations dynamically. Adaptive weight adjustments maintained a balance between exploration and exploitation, ensuring efficient route optimization. For Particle Swarm Optimization (PSO), velocity updates were adapted for VRP's discrete nature. Instead of continuous adjustments, swaps and shifts between customer locations defined the particle updates, preserving solution feasibility. A customized cost function evaluated solutions based on total travel cost while penalizing infeasible assignments. Dynamic penalty adjustments prioritized feasible solutions, ensuring adherence to vehicle capacity constraints. To enable robust comparisons, EHO and PSO were systematically tuned by optimizing population size, iteration limits, and weight coefficients.

5.2.1 Observations and Analysis

Experimental results indicate that **EHO outperformed PSO** in solving VRP, achieving **lower total costs** while maintaining computational efficiency. EHO's structured herding behavior allowed for **adaptive refinement of vehicle assignments**, leading to more balanced and efficient route distributions. In a test case with 20 customers and 5 vehicles, EHO achieved a total cost of **523.85**, compared to **693.38** for PSO. Additionally, EHO demonstrated a significantly lower execution time, reinforcing its viability for large-scale VRP instances.

The route visualizations in Figures 5.1 and 5.2 illustrate the optimization differences between EHO and PSO. The EHO-generated routes exhibit shorter travel distances and improved customer allocation, whereas the PSO-generated routes display longer paths and inefficient vehicle utilization.

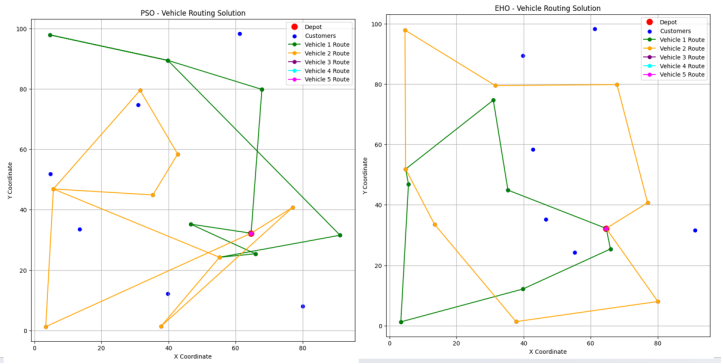


Figure 2. VRP Route Optimization using PSO and EHO

Further analysis is shown in **Figure 5.3**, where the bar chart compares the best solutions obtained by EHO and PSO, demonstrating a consistent cost reduction using EHO.

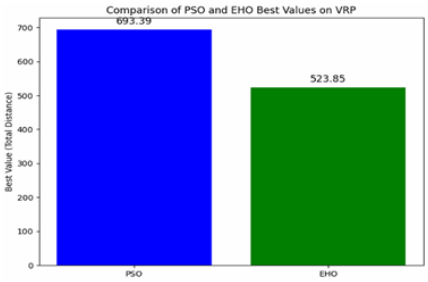


Figure 3. Comparison of Best Route Costs for PSO and EHO on VRP

These findings highlight that **EHO's structured herding approach significantly enhances VRP optimization**, outperforming traditional heuristics like PSO. By integrating **adaptive control mechanisms and specialized cost functions**, EHO efficiently manages **multi-vehicle constraints**, producing **high-quality solutions within computationally feasible limits**.

6. Benchmarking EHO on complex functions

The efficacy of Elephant Herding Optimization (EHO) was further evaluated on complex multimodal benchmark functions, which pose challenges due to the presence of multiple local minima. A crucial aspect of benchmarking was to test EHO's adaptability, convergence behavior, and ability to escape deceptive local optima. The first function considered is the Rastrigin function, which is highly multimodal and widely used in optimization research.

6.1 Rastrigin Function

The **Rastrigin function** is a widely used benchmark in continuous optimization, featuring a highly multimodal landscape with numerous local minima. Its periodic structure challenges algorithms to balance **exploration and exploitation**. The function is defined as:

$$f(X) = An + \sum_{i=1}^{n} \left[x_i^2 - A\cos(2\pi x_i)\right], \quad A = 10, \quad x_i \in [-5.12, 5.12]$$

where n is the dimensionality, x_i the i-th component of the solution, and A controls amplitude. The global minimum occurs at $f(X) = 0$ when $X = [0, 0, \ldots, 0]$. Algorithms must efficiently escape local minima while converging to the global optimum due to its periodic structure.

6.1.1 Implementation of Elephant Herding Optimization (EHO)

The EHO algorithm was used to optimize the Rastrigin function, leveraging its structured search mechanism. Elephants were randomly initialized within the search space, and fitness was evaluated based on function values. The herding behavior allowed elephants to collaborate in local search while maintaining global diversity. Matriarch-led updates guided elephants toward promising regions, preventing premature convergence, while male separation reintroduced diversity by repositioning trapped individuals. An adaptive mutation strategy gradually reduced mutation intensity, ensuring early exploration and late-stage refinement for improved convergence.

6.1.2 Comparative Analysis with Particle Swarm Optimization (PSO)

To benchmark performance, EHO was compared with PSO, which updates particle velocities based on individual and global best solutions. While PSO converges rapidly, it often suffers from premature stagnation, struggling to escape local minima. In contrast, EHO's structured updates and dynamic mutation strategy provide a better balance between exploration and exploitation, preventing convergence to suboptimal solutions—a common issue in PSO's fixed learning parameters.

6.1.3 Graphical Analysis of Convergence

A convergence study was conducted to analyze the behavior of EHO and PSO in solving the Rastrigin function. **Table 3** presents the fitness values recorded at key iterations, highlighting EHO's superior convergence characteristics.

Iteration	PSO Best Value	EHO Best Value
1	117.7485	88.4440
200	55.6717	21.2491
400	46.6527	21.2491
600	44.9307	21.2491
800	44.8767	16.9581
1000	44.8595	16.1309

Table 3. Convergence Values of PSO vs. EHO on Rastrigin

Figure 4 illustrates the convergence trajectories, comparing their ability to reach the global minimum efficiently.

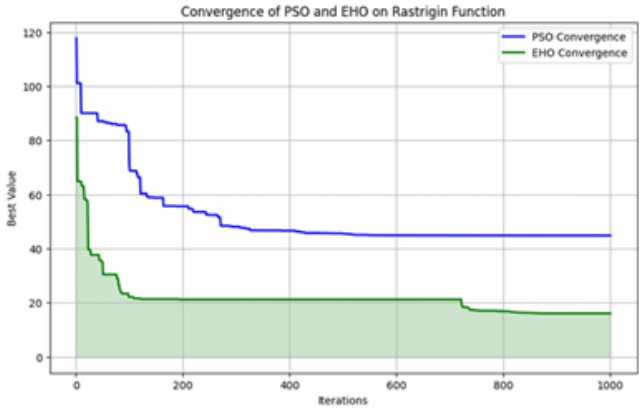


Figure 4. Convergence of EHO vs. PSO on Rastrigin Function

6.1.4 Observations and Insights

Experimental results confirm that EHO significantly outperforms PSO in optimizing the Rastrigin function, achieving lower fitness values while demonstrating greater robustness in multimodal search spaces. The structured herding behavior of EHO enables progressive refinement, preventing stagnation in local minima and leading to more efficient global convergence. Unlike PSO, which exhibits oscillatory behavior and premature stagnation, EHO maintains a steady decline in fitness, ensuring consistent solution quality. These results establish EHO as a superior alternative for solving highly multimodal, deceptive landscapes in continuous optimization.

6.2 Ackley Function

The Ackley function is a widely used optimization benchmark with multiple local minima and a single global minimum at the origin. It is defined as:

$$f(x) = -a\exp\left(-b\sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2}\right) - \exp\left(\frac{1}{d}\sum_{i=1}^{d}\cos(cx_i)\right) + a + \exp(1) \tag{1}$$

where $a = 20$, $b = 0.2$, and $c = 2\pi$. The function's exponential decay and oscillatory nature make it a challenging test for optimization algorithms, requiring an effective balance between **exploration and exploitation**.

6.2.1 Implementation of EHO

To optimize the Ackley function, EHO used a structured population-based search, initializing the herd randomly and evaluating fitness based on function values. Cooperative herding enabled local exploitation while preserving global diversity in the multimodal landscape. Matriarch-led updates directed the search toward optimal regions, preventing premature convergence, while male separation reintroduced diversity. A gradually decreasing mutation intensity ensured a smooth transition from exploration to exploitation, facilitating efficient convergence to the global minimum.

6.2.2 Execution and Results

The results show that EHO achieved the lowest function value (4.80) among all algorithms, demonstrating its effectiveness in handling the Ackley function. PSO followed closely (5.09), while GA (15.81) and ACO (19.94) performed significantly worse.

Algorithm	Best Value	Execution Time (s)
EHO	**4.80**	**0.21845**
PSO	5.09	0.11082
GA	15.81	0.09717
ACO	19.94	0.52352

Table 4. Execution Performance on Ackley Function

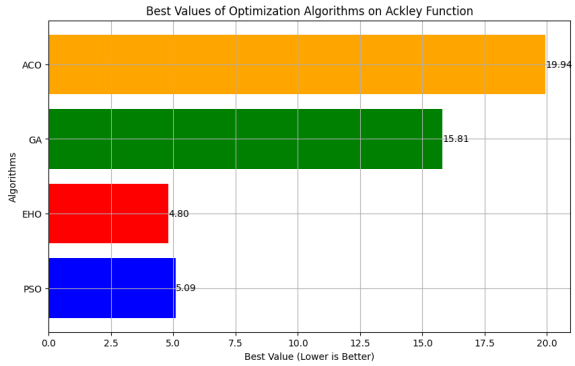


Figure 5. Execution time for Ackley Function

Although EHO demonstrated a slightly higher execution time (0.21845s), its superior optimization performance makes it well-suited for complex search spaces. ACO, with the highest execution time (0.52352s), was the least efficient.

7. EHO for Medical Image Registration

Medical image registration plays a crucial role in computational medical imaging, ensuring spatial alignment between images obtained from different modalities, time points, or perspectives.

7.1 Definition of Medical Image Registration and Its Challenges

Medical image registration aligns images from different modalities, time points, or perspectives, essential for disease monitoring, multi-modal fusion, and surgical comparisons. It estimates a transformation function to minimize discrepancies between a fixed reference and a moving image, ensuring anatomical consistency. However, challenges arise from biological variability, motion-induced deformations, and modality-specific differences in contrast, resolution, and acquisition parameters, making accurate registration computationally demanding. Traditional rigid registration uses intensity-based measures like mutual information or cross-correlation but struggles with structural differences. Deformable registration models complex transformations but faces challenges from high-dimensional spaces and local minima. Robust optimization is crucial for efficient, reliable alignment, especially amid noise and artifacts.

7.2 The Impact of Random Noise on Medical Image Registration

Noise is an inherent issue in medical imaging, arising from acquisition protocols, hardware limitations, and patient motion. It disrupts pixel intensities and structural coherence, complicating feature extraction and misguiding optimization algorithms into converging at incorrect local optima. Mathematically, it is modeled as an additive Gaussian perturbation, causing intensity variations that distort similarity measures like normalized mutual information and mean squared error (MSE), reducing registration accuracy. As noise inflates the variance of the dissimilarity function, by introducing spurious local minima, noise complicates optimization, particularly in high-dimensional tasks where distinguishing true anatomical structures from artifacts becomes challenging. Addressing these issues requires an optimization framework that balances exploration and exploitation while mitigating noise effects for robust image alignment.

7.3 Mathematical Representation of Medical Image Registration

Medical image registration determines an optimal transformation T that maps a moving image $I_m(x)$ onto a fixed reference image $I_f(x)$, formulated as:

$$T^* = \arg\min_T D(I_f(x), I_m(T(x))) + R(T) \tag{2}$$

where D quantifies misalignment, and $R(T)$ enforces smoothness constraints. The transformation can be **rigid, affine, or nonlinear**, depending on the application. A commonly used dissimilarity metric, **mean squared error (MSE)**, is given by:

$$D_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(I_f(x_i) - I_m(T(x_i)))^2 \tag{3}$$

where N is the total number of pixels. Minimizing D_{MSE} improves structural consistency and reduces intensity mismatches, while $R(T)$ prevents implausible deformations by ensuring transformation smoothness.

7.4 Modeling Noise Effects in Medical Image Registration

Noise is introduced as a stochastic perturbation in pixel intensities and is typically modeled as:

$$I_n(x) = I(x) + \eta(x)$$

where $\eta(x) \sim N(0, \sigma^2)$ represents Gaussian noise with a mean of zero and variance σ^2, controlling the noise intensity. This modification impacts the similarity function, altering the MSE measure as follows:

$$D_{\text{MSE, noise}} = \frac{1}{N}\sum_{i=1}^{N}(I_f(x_i) - (I_m(T(x_i)) + \eta(x_i)))^2$$

Noise increases error terms, distorts optimization landscapes, and raises the risk of suboptimal convergence. Integrating noise modeling into registration allows systematic evaluation of optimization robustness under real-world conditions.

7.5 Framework for Applying EHO in Medical Image Registration

Elephant Herding Optimization (EHO) optimized medical image registration by encoding transformation parameters (translation, rotation, scaling) within a multidimensional space. Each elephant represented a candidate transformation, evaluated using Mean Squared Error (MSE). The population was initialized for diversity and divided into herds, with leaders guiding cooperative and competitive updates—aligning elephants with their leader while pushing them toward the global best solution. Adaptive mutation and perturbation prevented premature convergence and improved resilience against noise, ensuring robust exploration and optimal alignment.

7.5.1 Adaptations for Noise-Induced Challenges

Medical image registration is affected by noise, distorting similarity metrics and introducing local minima. To counter this, EHO incorporated noise-robust fitness evaluations, smoothing fitness values for better leader selection. Adaptive mutation introduced controlled perturbations, preventing stagnation, while reinitialization preserved diversity, ensuring effective search space exploration. These adaptations enhanced EHO's robustness in noisy conditions.

7.5.2 Implementation of Comparative Algorithms (PSO, GA, ACO)

To benchmark EHO, we implemented PSO, GA, and ACO. PSO modeled transformations as particles, adjusting via inertia-weighted velocity updates. GA evolved transformations as chromosomes through crossover, mutation, and fitness smoothing for noise robustness. ACO optimized transformations as graph nodes using pheromone deposition and dynamic evaporation to mitigate noise. These implementations enabled a comparative evaluation of EHO's effectiveness.

```
Testing PSO on Medical Image Registration...
PSO Best Value (MSE): 0.133565, Execution Time: 4.93734 seconds
PSO Best Position: Angle=0.95, Tx=-20.00, Ty=-2.94

Testing EHO on Medical Image Registration...
EHO Best Value (MSE): 0.135149, Execution Time: 1.67352 seconds
EHO Best Position: Angle=-0.88, Tx=4.94, Ty=5.35

Testing GA on Medical Image Registration...
GA Best Value (MSE): 0.134779, Execution Time: 3.27865 seconds
GA Best Position: Angle=-3.04, Tx=-17.32, Ty=4.04

Testing ACO on Medical Image Registration...
ACO Best Value (MSE): 0.138497, Execution Time: 1.26770 seconds
ACO Best Position: Angle=3.26, Tx=20.00, Ty=20.00

===== Summary of Results =====
PSO: Best MSE = 0.133565, Time = 4.93734 seconds
EHO: Best MSE = 0.135149, Time = 1.67352 seconds
GA:  Best MSE = 0.134779, Time = 3.27865 seconds
ACO: Best MSE = 0.138497, Time = 1.26770 seconds
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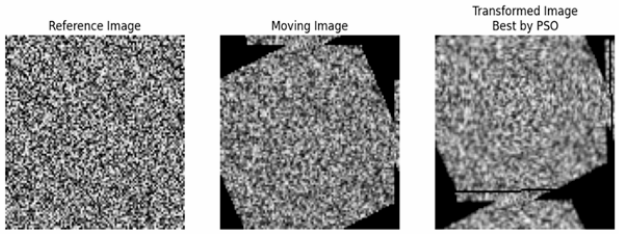


Figure 6. Medical Image Registration Results: Reference, Moving, and Transformed Image (PSO)

7.6 Execution Time, MSE, and Scalability Comparison

This section compares EHO, PSO, GA, and ACO in terms of execution time, Mean Squared Error (MSE), and scalability, evaluating their efficiency and accuracy in medical image registration.

7.6.1 Execution Time Analysis

Execution time is crucial in medical imaging for efficient clinical workflows. Figure 6.2 shows ACO had the fastest execution (1.26775s) due to its probabilistic transitions, followed by EHO (1.67355s) with its balanced herd-based search. GA (3.27865s) and PSO (4.93735s) were slower due to iterative genetic operations and velocity-based updates. PSO's reliance on best positions increased processing overhead, while GA's crossover and mutation added complexity. These results highlight EHO's efficiency for real-time image registration.

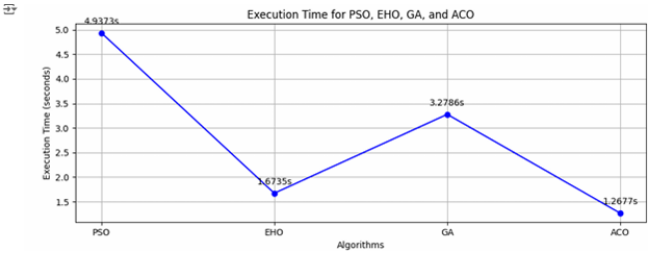


Figure 7. Execution Time for PSO, EHO, GA, and ACO

7.6.2 MSE-Based Accuracy Evaluation

MSE measures registration accuracy, with lower values indicating better alignment. Figure 6.3 shows PSO achieved the lowest MSE (0.133565) but with high computational cost. EHO (0.135149) demonstrated superior accuracy while maintaining efficiency. GA (0.134779) showed moderate performance, while ACO (0.138497) performed the worst, struggling with continuous optimization. EHO's balance of accuracy and speed makes it a strong choice for large-scale applications.

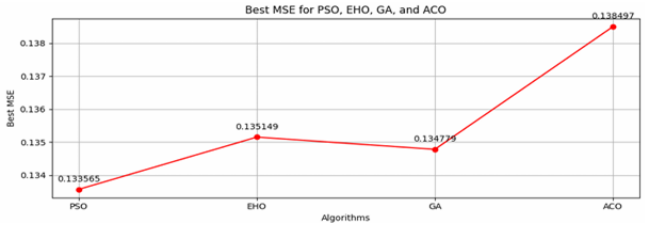


Figure 8. Best MSE for PSO, EHO, GA, and ACO

7.6.3 Scalability Testing

Scalability is essential for large-scale medical imaging applications. As **Figure 6.4** illustrates, EHO efficiently scales across increasing dataset sizes while maintaining performance. For a **50-image dataset**, EHO processed data in **17 seconds**, significantly outperforming PSO (**68s**) and GA (**60s**). EHO's hierarchical structure enabled efficient parallel evaluations, reducing computational overhead. ACO, though initially efficient, struggled with large datasets, exceeding **50s**. These results confirm EHO's adaptability, making it a viable choice for large-scale image registration.

```
Running experiment with 10 images...
Running experiment with 20 images...
Running experiment with 30 images...
Running experiment with 40 images...
Running experiment with 50 images...
```

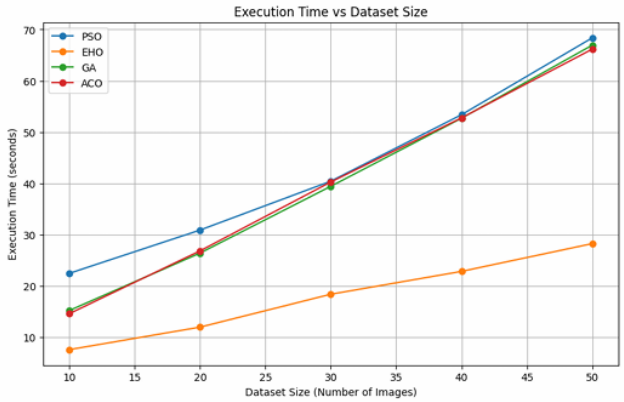


Figure 9. Execution Time vs. Dataset Size for PSO, EHO, GA, and ACO

7.7 Overall Insights

The evaluation highlights EHO's efficiency and accuracy in medical image registration, making it ideal for large-scale tasks. EHO achieved fast execution with high accuracy, outperforming ACO in accuracy and PSO in efficiency. While PSO had the best accuracy, its high execution time limits real-time use. GA showed moderate performance but was computationally expensive, and ACO's high MSE reduced its applicability. These findings establish EHO as a robust, scalable, and efficient solution for medical image registration.

7.8 Applications and Implications

EHO enhances large-scale medical imaging by achieving low execution times and high accuracy, ensuring applicability in real-time workflows, population studies, and resource-limited settings. Its robustness to noise makes it ideal for artifact-prone medical data. Beyond registration, EHO

supports MRI slice enhancement and 3D reconstruction, improving disease localization and surgical planning. These findings position EHO as a key tool for AI-driven radiology, automated diagnostics, and advanced healthcare solutions.

8. EHO for MRI Image Enhancement and 3D Reconstruction

Magnetic Resonance Imaging is essential in modern diagnostics, but raw outputs often suffer from noise, low contrast, and motion artifacts, affecting reliability.

8.1 Role of MRI Enhancement in Diagnostics and 3D Modeling

MRI provides high-resolution soft tissue imaging but suffers from noise, low contrast, and motion artifacts, reducing diagnostic reliability. Enhancement techniques refine images for better pathology detection, crucial in oncology where tumor boundary clarity impacts clinical decisions. Traditional methods like histogram equalization and Gaussian filtering often lead to over-smoothing or excessive contrast. In contrast, EHO dynamically enhances pixel intensities, balancing contrast adjustments while preserving critical details for accurate diagnosis.

8.1.1 Need for 3D Reconstruction in Comprehensive Analysis

2D MRI slices lack spatial continuity, making 3D reconstruction essential for accurate visualization in neurosurgery. Traditional interpolation causes artifacts, distorting spatial relationships. EHO-driven optimization preserves structural details, enhancing diagnostic accuracy and AI-driven tumor classification.

8.1.2 Overview of the BRATS Dataset and Its Relevance

The BRATS dataset, with multimodal MRI scans and tumor annotations, is ideal for benchmarking MRI enhancement and 3D reconstruction. Its intensity variations and noise artifacts simulate real-world challenges. EHO's performance on BRATS validates its ability to refine intensity distributions, suppress noise, and enhance tumor visualization.

8.2 Methodology for MRI Slice Enhancement

MRI slice enhancement is crucial for improving diagnostic accuracy and enabling robust 3D reconstruction. This section details enhancement methodologies using EHO, PSO, ACO, and GA, each tailored to address noise suppression, contrast optimization, and structural preservation.

8.2.1 Algorithmic Approach for MRI Enhancement Using EHO

EHO optimized MRI enhancement by modeling pixel intensity refinement as an optimization problem. Each elephant represented an enhancement configuration balancing contrast and noise suppression. The initialization phase generated intensity matrices within the 0–255 grayscale range for diverse exploration. Iterative updates followed:

$$X_{\text{new}} = X_{\text{current}} + r \cdot (X_{\text{herd-best}} - X_{\text{global-best}}) \tag{4}$$

where r regulates search intensity. Final images were normalized via percentile-based rescaling, ensuring clinically relevant contrast. EHO's adaptive strategy outperformed fixed-intensity methods, preserving fine anatomical details while preventing excessive contrast distortions.

8.2.2 Adaptations for MRI Enhancement Using PSO

Particle Swarm Optimization (PSO) modeled pixel intensity adjustments as particles refining enhancement solutions. The velocity update followed:

$$V_{\text{new}} = \omega V_{\text{old}} + c_1 r_1 (P_{\text{best}} - X_{\text{current}}) + c_2 r_2 (G_{\text{best}} - X_{\text{current}}) \tag{5}$$

where ω is inertia, and c_1, c_2 regulate cognitive and social influence. PSO's reliance on global best solutions enabled fast convergence but struggled with noise, leading to suboptimal enhancements in MRI slices with high contrast variability.

8.2.3 Implementation of MRI Enhancement Using ACO

Ant Colony Optimization (ACO) modeled MRI enhancement as a graph traversal problem, where pixel intensity adjustments followed probabilistic paths guided by pheromone deposition:

$$P_{ij} = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_k \tau_{ik}^{\alpha} \cdot \eta_{ik}^{\beta}} \tag{6}$$

where pheromone intensity (τ_{ij}) and heuristic information (η_{ij}) influenced path selection. Pheromone evaporation prevented stagnation, while heuristic weighting preserved edges. Though ACO enhanced contrast, its iterative pheromone updates increased computational overhead compared to EHO and PSO.

8.2.4 Comparative Methodology for MRI Enhancement Using GA

Genetic Algorithm (GA) modeled MRI enhancement as an evolutionary process, encoding intensity refinements as chromosomes optimized via selection, crossover, and mutation. Selective introduction of Gaussian noise prevented premature convergence, while elitism preserved high-performing solutions. A dynamic mutation rate balanced exploration and exploitation. Though robust, GA converged significantly slower than EHO and PSO, especially in noisy slices. Histogram-equalized initialization improved performance, reducing noise susceptibility.

8.3 Results and Analysis for MRI Slice Enhancement

The evaluation of MRI slice enhancement relies on two fundamental metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). These measures assess the effectiveness of each algorithm in reducing noise while preserving critical anatomical details.

8.3.1 PSNR and SSIM Benchmark Analysis for MRI Enhancement Algorithms

Peak Signal-to-Noise Ratio (PSNR) measures noise suppression, with higher values indicating better enhancement, while Structural Similarity Index (SSIM) quantifies perceptual similarity, approaching 1 for optimal structural integrity. **PSNR** is computed as:

$$PSNR = 10 \cdot \log_{10} \left(\frac{L^2}{MSE} \right) \tag{7}$$

where L is the maximum pixel intensity (255 for 8-bit images), and MSE is given by:

$$MSE = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} (I(i,j) - K(i,j))^2 \tag{8}$$

where $I(i,j)$ and $K(i,j)$ denote pixel intensities of the original and enhanced images, respectively. **SSIM**, which evaluates structural similarity, is defined as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{9}$$

where μ_x, μ_y represent mean pixel intensities, σ_x^2, σ_y^2 denote variances, and σ_{xy} is the covariance between images. Constants C_1 and C_2 stabilize computations.

The PSNR and SSIM scores for EHO, PSO, and GA are summarized in **Figure 10**. GA demonstrated the highest PSNR (14.33) and SSIM (0.89), benefiting from evolutionary mechanisms that iteratively refine pixel intensity distributions. EHO and PSO achieved competitive PSNR values (13.72) but exhibited lower SSIM (0.67), indicating a balanced approach to noise reduction and contrast preservation.

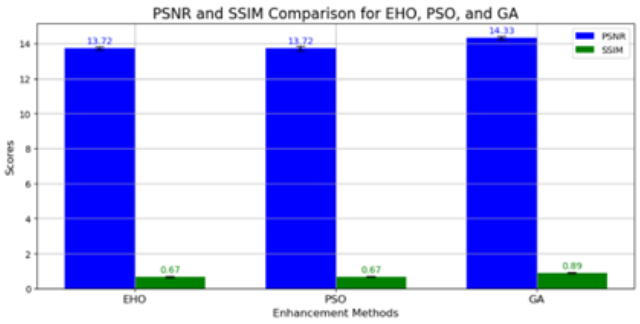


Figure 10. PSNR and SSIM Comparison for EHO, PSO, and GA

8.3.2 Execution Time Analysis Across Algorithms

Execution time is critical in real-time MRI applications. **Figure 11** compares the execution times of EHO, GA, and PSO. EHO was the fastest, enhancing each slice in **0.3004s**, leveraging herd-based cooperation for rapid convergence. GA, though achieving better PSNR and SSIM, took **0.3948s** due to its iterative evolutionary process. PSO was the slowest (**0.6414s**), hindered by velocity adjustments and global-best dependencies. These results establish EHO as an efficient approach for large-scale medical imaging, balancing enhancement quality and computational speed.

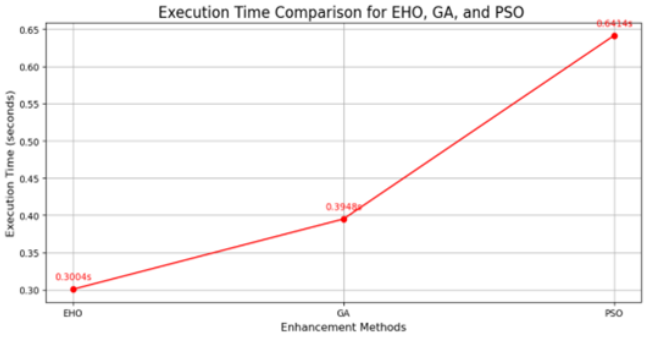


Figure 11. Execution Time Comparison for EHO, GA, and PSO

8.3.3 Results of MRI Enhancement by All Approaches

Figure 12 compares MRI slices enhanced using EHO, PSO, and GA. The original noisy, low-contrast slice was processed for clarity and artifact suppression. EHO preserved structural details while enhancing contrast efficiently. PSO achieved similar contrast but required longer execution times and occasionally lost finer structures. GA produced the most refined output with superior noise reduction but incurred higher computational costs. EHO balances speed and quality, making it ideal for real-time imaging, while the accuracy-time trade-off underscores the need for task-specific optimization.

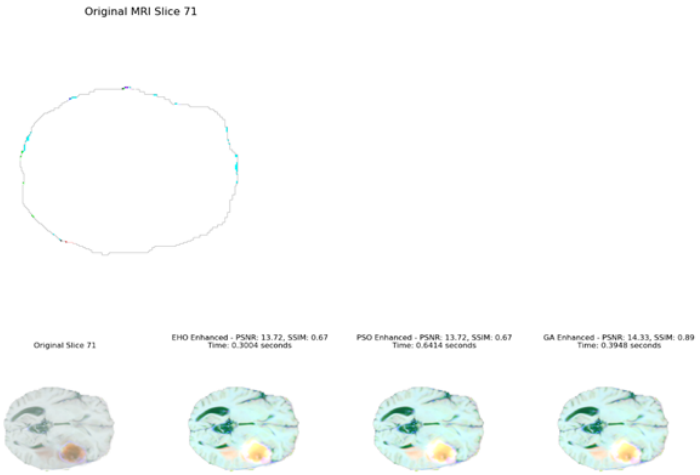


Figure 12. Comparison of Original and Enhanced MRI Slices for EHO, PSO, and GA

8.4 Individual 3D Model Creation for Each Enhanced MRI Slice

The reconstruction of 3D models from enhanced MRI slices is an essential step in medical imaging, enabling the transition from 2D cross-sectional scans to volumetric representations. This process enhances spatial comprehension, particularly in tumor analysis and surgical planning.

8.4.1 Methodology for Slice-by-Slice 3D Reconstruction

The methodology employs EHO, PSO, GA, and ACO, to individual MRI slices which are enhanced, aligned and processed to generate accurate 3D reconstructions, preserving critical anatomical features. The Marching Cubes algorithm from the `skimage` library was employed to identify isosurfaces based on intensity transitions, creating a triangular mesh that delineates structural boundaries which were visualised using the `mayavi` library, facilitating detailed rendering and interactive exploration. Ensuring inter-slice continuity prevented artifacts, preserving the accurate spatial structure of brain tissues.

8.4.2 Results from Slice-by-Slice 3D Models with Enhanced MRI Slices

The results from the slice-by-slice reconstruction demonstrated that EHO-based enhancement led to smoother and more precise 3D models, with well-defined anatomical contours. Compared to PSO and GA, EHO minimized voxel inconsistencies and preserved finer details, reducing artifacts. This capability is particularly valuable in applications requiring precise localization of pathological structures, such as neurosurgical planning.



Figure 13. 3D Model of an Individual MRI Slice Based on User Selection

8.5 Complete 3D Model Creation from All Slices

The full 3D brain reconstruction process aggregates all enhanced MRI slices into a single volumetric model, providing a comprehensive view of anatomical structures. Unlike the individual slice-based approach, this method demands meticulous alignment of slices to preserve inter-slice continuity.

8.5.1 Framework for Full 3D Brain Reconstruction Using Enhanced Slices

Enhanced slices processed through bio-inspired algorithms ensure that structural integrity and contrast details are retained, making the volumetric model more suitable for diagnostic and analytical applications.

3D reconstruction is TSP-reducible, where MRI slices act as nodes, and the goal is to optimize traversal for anatomical continuity. The cost function, based on inter-slice similarity metrics, ensures minimal distortion. Mapping this to TSP allows EHO to optimize slice ordering, reducing misalignment and enhancing reconstruction fidelity.

The reconstruction pipeline employs sequential stacking of enhanced slices, followed by Marching Cubes-based iso-surface extraction to generate a 3D triangular mesh. Post-processing techniques, such as Gaussian smoothing, are applied to refine surface details. The use of enhanced slices significantly improves the quality of the reconstructed model, mitigating noise artifacts and ensuring clearer differentiation of normal and pathological tissues.

8.5.2 Results and Observations: 3D Brain Reconstruction and Tumor Detection

The final 3D models constructed from all slices revealed the superior performance of EHO in producing anatomically accurate representations. The method effectively maintained smooth surfaces and well-defined structural boundaries while minimizing artifacts. PSO-based reconstructions, while computationally efficient, introduced minor distortions due to premature convergence. GA yielded high-contrast models but required longer processing times, making it less viable for real-time applications.

Tumor segmentation was successfully integrated into the reconstructed models, enabling the identification of abnormal regions with high precision. The enhanced 3D visualization of tumors facilitates preoperative assessment, allowing clinicians to analyze tumor size and spatial relationships with surrounding structures. These findings highlight the potential of bio-inspired enhancement techniques in advancing volumetric medical imaging, offering a robust framework for both diagnostic and interventional applications.

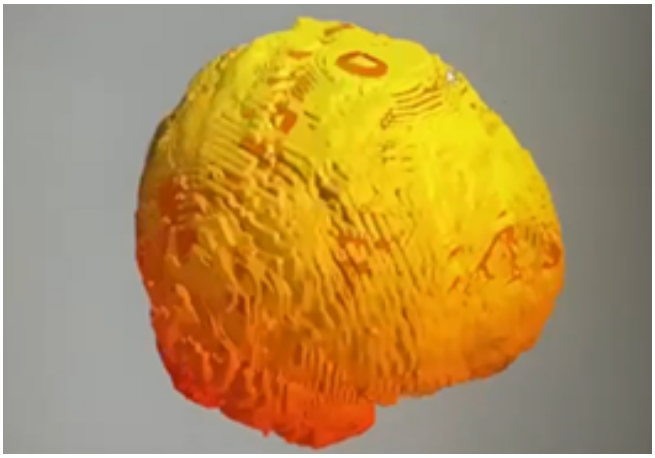


Figure 14. Complete 3D Brain Model Constructed from Enhanced MRI Slices

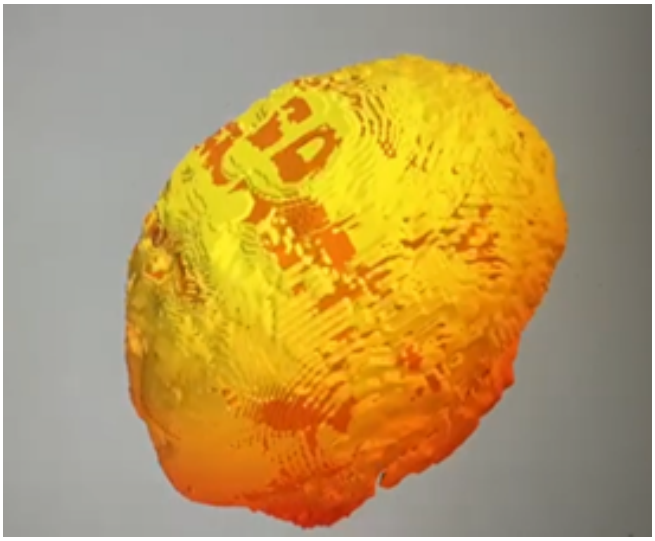


Figure 15. Final 3D Brain Model Showing Structural Details and Tumor Regions

8.6 Medical Benefits of 3D Models

The transition from enhanced 2D MRI slices to comprehensive 3D brain models marks a significant advancement in diagnostic imaging. Reconstructed 3D models, particularly those generated using bio-inspired algorithms such as EHO, provide superior visualization for assessing tumor size, location, and spatial distribution.

8.6.1 Enhanced Visualization for Accurate Tumor Diagnosis

Traditional 2D imaging may fail to fully capture tumors and their interaction with critical structures. 3D reconstruction integrates enhanced slices, providing a comprehensive view for precise tumor detection and evaluation.

EHO's optimization preserves contrast and intensity variations, crucial for delineating tumors near critical areas. Intensity-based segmentation differentiates normal and pathological tissues, while interactive 3D visualization aids clinical decision-making, enhancing preoperative planning and treatment assessment.

8.6.2 Scalability for Real-Time MRI Enhancements and Broader Applications

Real-time MRI enhancement and 3D reconstruction are vital for intraoperative imaging and emergency care. EHO's efficiency enables fast, high-quality processing, supporting surgical guidance and diagnostics.

Beyond brain imaging, these techniques extend to CT, PET, and SPECT scans, aiding radiation therapy planning and treatment monitoring. Integrating bio-inspired optimization with AI enhances predictive modeling, abnormality detection, and automated diagnostics, advancing real-time medical imaging and clinical decision-making.

7. Conclusion

Real-time MRI enhancement and 3D reconstruction are vital for intraoperative imaging and emergency care. EHO's efficiency enables fast, high-quality processing, supporting surgical guidance and diagnostics. Beyond brain imaging, these techniques extend to CT, PET, and SPECT scans, aiding radiation therapy planning and treatment monitoring. Integrating bio-inspired optimization with AI enhances predictive modeling, abnormality detection, and automated diagnostics, advancing real-time medical imag-

ing and clinical decision-making.

8. Future Directions for Real-Time Applications

Future research should optimize these methodologies for real-time applications like intraoperative MRI and emergency diagnostics by enhancing computational efficiency. Parallelizing EHO on GPU architectures could accelerate processing, making real-time imaging feasible. Integrating machine learning with bio-inspired optimization could refine tumor segmentation and anomaly detection, improving diagnostic automation.

Expanding to large-scale, multi-modal datasets will ensure broader clinical adoption, while cloud-based integration could facilitate remote diagnostics and collaborative healthcare. Refining these techniques with emerging technologies will further establish bio-inspired optimization as a key driver in real-time medical imaging.

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